

01. Cancer Diagnosis

Cancer diagnosis is the process of identifying the presence, type, and extent of cancer. Early diagnosis significantly improves treatment success and survival rates.

Doctors use a combination of medical history, physical examination, laboratory tests, imaging studies, and biopsy to confirm the diagnosis.

02. Medical History

The physician asks about:

- Current symptoms
- Duration of symptoms
- Family history of cancer
- Smoking history
- Alcohol consumption
- Previous illnesses
- Occupational exposure
- Lifestyle habits

This information helps determine possible cancer risk factors.

03. Physical Examination

A complete physical examination includes checking for:

- Lumps or swelling
- Enlarged lymph nodes
- Skin abnormalities
- Organ enlargement
- Weight loss
- Abnormal bleeding
- Painful areas

Physical findings guide further investigations.

04. Laboratory Tests

Common laboratory tests include:

Complete Blood Count (CBC)

Detects abnormalities in:

- Red blood cells
- White blood cells
- Platelets

Useful for leukemia and lymphoma.

Blood Chemistry Tests

Measures:

- Liver function
- Kidney function
- Blood sugar
- Electrolytes

Helps evaluate overall health before treatment.

Tumor Marker Tests

Certain cancers produce proteins called tumor markers.

Examples include:

- PSA for prostate cancer
- CA-125 for ovarian cancer
- AFP for liver cancer
- CEA for colorectal cancer

Tumor markers are not sufficient alone to diagnose cancer.

05. Imaging Tests

Imaging helps locate tumors and determine whether cancer has spread.

Common imaging methods include:

X-Ray

Used for:

- Lung cancer
- Bone cancer
- Chest abnormalities

Ultrasound

Useful for:

- **Liver**
 - **Thyroid**
 - **Breast**
 - **Abdomen**
 - **Pelvis**
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CT Scan

Provides detailed cross-sectional images.

Commonly used to detect:

- **Lung cancer**
 - **Liver cancer**
 - **Colon cancer**
 - **Pancreatic cancer**
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MRI Scan

Produces high-resolution images.

Helpful for:

- **Brain tumors**
 - **Spine tumors**
 - **Soft tissue cancers**
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PET Scan

Detects areas of increased metabolic activity.

Useful for:

- **Cancer staging**
 - **Detecting metastasis**
 - **Monitoring treatment response**
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06. Biopsy

A biopsy is the gold standard for cancer diagnosis.

A small tissue sample is removed and examined under a microscope.

Types include:

- **Needle biopsy**
 - **Surgical biopsy**
 - **Endoscopic biopsy**
 - **Bone marrow biopsy**
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07. Cancer Staging

Staging describes:

- **Tumor size**
- **Lymph node involvement**
- **Spread to other organs**

Accurate staging helps determine the best treatment.

08. TNM Staging System

T (Tumor)

Describes the size of the primary tumor.

Example:

- **T1**
 - **T2**
 - **T3**
 - **T4**
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N (Nodes)

Shows whether nearby lymph nodes contain cancer.

Examples:

- **N0**
- **N1**
- **N2**

- N3
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M (Metastasis)

Indicates whether cancer has spread.

Examples:

- M0 – No distant spread
 - M1 – Cancer has spread
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09. Overall Cancer Stages

Stage 0

Cancer cells remain confined to their original location.

Treatment success is very high.

Stage I

Small localized tumor.

No lymph node involvement.

Usually highly curable.

Stage II

Larger tumor with limited local spread.

Treatment may involve surgery and chemotherapy.

Stage III

Cancer has spread to nearby lymph nodes.

Requires multimodal treatment.

Stage IV

Cancer has spread to distant organs.

Treatment focuses on controlling disease and improving quality of life.

10. Importance of Early Diagnosis

Early detection offers many benefits:

- **Higher survival rates**
 - **Less aggressive treatment**
 - **Lower treatment costs**
 - **Better quality of life**
 - **Reduced complications**
 - **Faster recovery**
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11. Follow-Up After Diagnosis

Patients should attend regular follow-up appointments to:

- **Monitor treatment response**
- **Detect recurrence early**
- **Manage treatment side effects**
- **Maintain overall health**